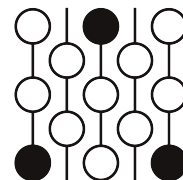


Review of Submission 2C

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The AI got the problem correct. This is a working proof, with all essential details included.

The proof in the first half works well, and it follows a clear outline. All details seem correct.

The proof in the second half seems to work, though the exposition seemed a bit heavy on calculations and light on conceptual explanation. In particular, there was a statement about C^1 -convergence of the approximation which needed to be interpreted generously. See inline comment below. I did not check the calculations line by line but it was clear to me that this approach works.

Novelty. Let me first discuss the context for the problem and then describe how the AI did. The problem is based on my solution of the Halpern-Weaver Conjecture, a result which asks for the infimal aspect ratio needed for a smooth and isometric embedding of a flat Moebius band into $\mathbb{R}^3$. My paper was published in the *Annals* in January, 2025 and there are also some of accounts of the result in the popular media. The problem the AI considered is the same problem in a different category of maps, piecewise linear maps with a certain side condition called squeezing.

The first half of the problem amounts to showing that $\sqrt{3}$ is a lower bound on the allowable aspect ratio. For this, the AI seems to have followed my paper very closely, adapting essentially every one of my arguments to the new category of mappings. While the AI did cite my paper (just as context for the second half) it did not acknowledge that it was following the arguments in my paper very closely. I know the AI read my paper because it used the same terminology and notation from my paper – e.g. “T-patterns” and “bends”, and the letters T, zb, D, H for the main objects in the geometric part of the proof.

The second half is not derived from my paper, and it essentially gives a nuts-and-bolts proof involving smooth approximation of an already known smooth construction that goes back to Michael Sadowsky in 1930. I didn’t really like this proof that much. It uses a black box, Sadowsky’s result, and then does PL approximation until it wins. So, this half is sort of novel, but it is what you would do if you did not really have any artistic sense.

2 Recommendation

I think that this paper is publishable with minor revisions. First, I think that the paper should make completely clear that it is getting the main ideas from my paper. Second, I think that the paper should clean up the second half. I think it is acceptable as a solution to use a smooth result as a black box, but it would be nice to have an explicit and self-contained exposition. I would not insist on this, however. The PL approximation approach here is not too bad – better than a similar argument that one of the other AIs did – so I wouldn't insist on a total overhaul. In case, the author really wants to use the PL approach, I might just try for a more conceptual argument that follows a similar outline.

3 Submission with inline comments

Abstract

We prove the stronger statement that a number $\beta > 0$ is realized in the sense of the question if and only if $\beta > \sqrt{3}$. Hence the infimum of the set of realized numbers is $\sqrt{3}$.

3.1 Preliminaries

Let

$$M_\beta = \Sigma / \langle G_\beta \rangle.$$

Thus M_β is the flat Mobius band obtained from the rectangle $[0, \beta] \times [0, 1]$ by identifying $(0, y) \sim (\beta, 1 - y)$. Its boundary is a circle of flat length 2β . We use the boundary coordinate

$$s(x, 0) = x \pmod{2\beta}, \quad s(x, 1) = x + \beta \pmod{2\beta}.$$

Lemma 1 (quotient and boundary contraction). *Suppose β is realized. Then the map f descends to a topological embedding*

$$\bar{f} : M_\beta \longrightarrow \mathbb{R}^3.$$

Moreover, the G_β -invariant clean triangulation descends to a finite clean triangulation of M_β , and every nondegenerate boundary arc $A \subset \partial M_\beta$ satisfies

$$\text{length}(\bar{f}(A)) < \text{length}_0(A),$$

where length_0 denotes flat boundary length.

Proof. Condition (2) gives $f \circ G_\beta = f$, so f descends to $\bar{f}$. The same condition says precisely that $\bar{f}$ is injective. Since M_β is compact and $\mathbb{R}^3$ is Hausdorff, $\bar{f}$ is an embedding.

A compact fundamental domain in Σ meets only finitely many triangles by local finiteness, so only finitely many G_β -orbits of triangles occur. Thus the quotient triangulation is finite.

Finally, on each boundary edge of the triangulation, f is affine and strictly shortens that edge. Hence it strictly shortens every subsegment of every boundary edge. Since every boundary arc is a finite union of such subsegments, its image has strictly smaller length. $\square$

Lemma 2 (the squeeze cone). *Let τ be a clean triangle with boundary edge AB and opposite vertex C . If an affine map $L : \tau \rightarrow \mathbb{R}^3$ decreases the length of AB and increases the lengths of AC and BC , then L increases the length of every segment joining C to a point of AB .*

Proof. Write $u = C - A$, $v = C - B$, so $u - v = B - A$. Let

$$q(w) = |Lw|^2 - |w|^2.$$

The hypotheses are

$$q(u) > 0, \quad q(v) > 0, \quad q(u - v) < 0.$$

For $0 \leq t \leq 1$,

$$q((1 - t)u + tv) = (1 - t)q(u) + tq(v) - t(1 - t)q(u - v) > 0.$$

Thus every cevian from C to the base AB is strictly lengthened. □

3.2 The canonical ruling and a T -pattern

For every clean triangle, foliate it by the straight segments joining the vertex opposite the boundary edge to points of the boundary edge. Adjacent clean triangles share a nonboundary edge, and that edge is one of the leaves in both triangles. Therefore these local foliations glue.

Lemma 3 (canonical ruling). *The quotient clean triangulation of M_β carries a continuous circle-family $\mathcal{L} = \{L_t\}_{t \in S^1}$ of straight clean segments with the following properties.*

- (i) *Each L_t is contained in a single clean triangle and joins the two boundary components in the universal cover.*
- (ii) *The relative interiors of distinct leaves are disjoint and the leaves cover M_β .*
- (iii) *The image $\bar{f}(L_t)$ is a straight segment in $\mathbb{R}^3$, and*

$$|\bar{f}(L_t)| > |L_t|_0 \geq 1.$$

- (iv) *A continuous orientation of the leaves reverses after one circuit of the parameter circle.*

Proof. Let $\mathcal{T}$ be the finite clean triangulation of M_β . Form the dual graph whose vertices are triangles and whose edges correspond to shared nonboundary edges. Every triangle has exactly two nonboundary edges, and every nonboundary edge is shared by two triangles, so the dual graph is 2-regular. It is connected: at each boundary vertex the link is an interval, so the triangles incident to that vertex are connected through shared nonboundary edges; since the triangulation is connected, the dual graph is connected. Hence the dual graph is a single cycle.

On a triangle τ , the family of cevians from the opposite vertex to the boundary edge is parametrized by an interval; its two endpoint leaves are the two nonboundary edges of τ . Gluing these intervals according to the cyclic dual graph gives a parameter circle S^1 . This proves the existence and continuity of the family $\{L_t\}$, and the relative interiors are plainly disjoint.

Each leaf is contained in a clean triangle, and by the preceding squeeze-cone lemma its image is strictly longer than its flat length. Since every clean segment crosses from $y = 0$ to $y = 1$ in the universal cover, its flat length is at least 1.

Finally, the leaf family is an interval-bundle structure on the Mobius band away from finitely many boundary vertices. Since M_β is the nontrivial I -bundle over a circle, the orientation line of the fibers is nontrivial. Equivalently, in the universal cover, going once around the leaf circle applies the deck transformation G_β , which exchanges the two boundary components. Hence any continuous orientation of the leaves is reversed after one circuit. $\square$

Reviewer 1

This is really the same as Lemma T in my paper on the optimal paper Moebius band. It is the main point.

Lemma 4 (T -pattern). *There are two distinct leaves L_s, L_t such that the two straight segments*

$$\bar{f}(L_s), \quad \bar{f}(L_t)$$

lie in perpendicular coplanar lines.

Proof. Lift the parameter circle to $\mathbb{R}$. Choose continuously oriented unit direction vectors $u(r)$ for the image segments and let $m(r)$ denote their midpoints. By the orientation-reversal property,

$$u(r + 2\pi) = -u(r), \quad m(r + 2\pi) = m(r).$$

Let

$$\Upsilon = \{(r, s) : r < s < r + 2\pi\} / (r, s) \sim (r + 2\pi, s + 2\pi).$$

Compactify Υ by adding two points: ∂_+ , corresponding to $s \rightarrow r$, and ∂_- , corresponding to $s \rightarrow r + 2\pi$. This compactification is a 2-sphere, and the swap involution

$$(r, s) \mapsto (s, r + 2\pi)$$

extends to the antipodal map on that sphere.

Define

$$F(r, s) = (u(r) \cdot u(s), (m(r) - m(s)) \cdot (u(r) \times u(s))).$$

Then F extends continuously to the compactification with

$$F(\partial_+) = (1, 0), \quad F(\partial_-) = (-1, 0),$$

and satisfies

$$F(s, r + 2\pi) = -F(r, s).$$

By the Borsuk–Ulam theorem, F has a zero. At such a zero, the two direction vectors are perpendicular, and the vector joining the midpoints is orthogonal to the normal $u(r) \times u(s)$. Therefore the two supporting lines are coplanar. This gives the desired pair of leaves. $\square$

3.3 The lower bound

This lemma is the main geometric construction. It is pretty much the same as what I have in my optimal paper Moebius band paper. Also, it could use a picture! For geometry solutions it would be very useful to have the AI do this.

Lemma 5 (geometry of a squeezed T -pattern). *If β is realized, then $\beta > \sqrt{3}$.*

Proof. Let T and B be the two leaves supplied by the preceding lemma. Write

$$T^* = \bar{f}(T), \quad B^* = \bar{f}(B).$$

Their supporting lines are perpendicular and coplanar. Interchange T and B , if necessary, so that the line supporting T^* does not meet the relative interior of B^* . Otherwise the two embedded segments would have an interior intersection.

Let u be the endpoint of B whose image is farthest from the line supporting T^* . Since B^* is perpendicular to that line and the line does not meet the relative interior of B^* , the distance h from $\bar{f}(u)$ to the line supporting T^* satisfies

$$h \geq |B^*| > |B|_0 \geq 1.$$

Let the endpoints of T divide ∂M_β into two boundary arcs H and D , with $u \in D$. Choose the signed horizontal displacement t of T so that

$$\text{length}_0(H) = \beta - t, \quad \text{length}_0(D) = \beta + t.$$

Then

$$|T|_0 = \sqrt{1 + t^2}.$$

Because T is a leaf, it is lengthened:

$$|T^*| > \sqrt{1 + t^2}.$$

Since H is a boundary path joining the endpoints of T ,

$$\sqrt{1 + t^2} < |T^*| \leq \text{length}(\bar{f}(H)) < \beta - t. \quad (1)$$

Now consider the planar triangle Δ whose base is T^* and whose third vertex is $\bar{f}(u)$. Its base has length $> |T|_0 = \sqrt{1 + t^2}$, and its height is $h > 1$. If $\vee$ denotes the union of the two nonbase sides of Δ , the elementary reflection argument gives

$$\text{length}(\vee) > \sqrt{5 + t^2}.$$

Indeed, among all triangles with base length b and height h , the sum of the two nonbase sides is at least $\sqrt{b^2 + 4h^2}$.

The boundary arc D runs from one endpoint of T to the other through u . Hence the two sides of $\vee$ are no longer than the two corresponding subarcs of $\bar{f}(D)$. Therefore

$$\sqrt{5 + t^2} < \text{length}(\vee) \leq \text{length}(\bar{f}(D)) < \beta + t. \quad (2)$$

From (1) and (2),

$$2\beta > \alpha(t) := \sqrt{1+t^2} + \sqrt{5+t^2},$$

and also

$$2\beta > \gamma(t) := 2\sqrt{5+t^2} - 2t.$$

Let $t_0 = 1/\sqrt{3}$. Then

$$\alpha(t_0) = \gamma(t_0) = 2\sqrt{3}.$$

Moreover, α is increasing for $t > 0$, and γ is strictly decreasing on $\mathbb{R}$. Thus for every real t ,

$$\max\{\alpha(t), \gamma(t)\} \geq 2\sqrt{3}.$$

Consequently $2\beta > 2\sqrt{3}$, so $\beta > \sqrt{3}$. □

3.4 Realization for every $\beta > \sqrt{3}$

Reviewer 1

The various citations here are rather confused. The best reference would be the existence of smooth isometric embeddings of aspect ratio converting to $\sqrt{3}$ using the *pseudofold* construction in the 1977 paper by Halpern and Weaver. My own paper doesn't really discuss this existence result at all, though it does sketch a proof of the existence of the bend foliation.

We use the classical existence theorem for paper Mobius bands: for every width $w > 0$ and every length λ with $\lambda/w > \sqrt{3}$, there exists a smooth embedded isometric Mobius band of width w and length λ , obtained by smoothing the triangular Mobius band. Moreover, it has a continuous ruling by straight bends whose preimages are straight segments joining the two boundary components. This is the standard Sadowsky–Wunderlich construction; see Sadowsky [1], the translation by Hinz–Fried [2], and the modern discussion in Schwartz [3].

Proposition 6. *Every $\beta > \sqrt{3}$ is realized.*

Reviewer 1

Right at the beginning of this proof it would be nice to discuss the strategy briefly, to explain the roles of w and λ .

Proof. Fix $\beta > \sqrt{3}$. Choose numbers w, λ such that

$$1 < w, \quad \sqrt{3}w < \lambda < \beta, \quad w^2 - 1 > \beta^2 - \lambda^2.$$

This is possible by taking λ sufficiently close to β and then $w > 1$ sufficiently close to 1.

Let $\rho = \lambda/\beta < 1$. Let

$$P_{\lambda,w} = ([0, \lambda] \times [0, w]) / (0, y) \sim (\lambda, w - y),$$

and let

$$I : P_{\lambda,w} \longrightarrow \mathbb{R}^3$$

be a smooth embedded paper Mobius band with a ruling by straight bends. Define

$$A : M_\beta \longrightarrow P_{\lambda,w}, \quad A[x, y] = [\rho x, wy].$$

This is well-defined because

$$A(x + \beta, 1 - y) = (\rho x + \lambda, w - wy),$$

which is the defining Mobius identification in $P_{\lambda,w}$. Put

$$F = I \circ A : M_\beta \longrightarrow \mathbb{R}^3.$$

Then F is an embedding.

Let $\{C_s\}_{s \in S^1}$ be the pulled-back bend ruling on M_β . Each C_s is a clean straight segment. If C_s has signed horizontal displacement d , with $|d| \leq \beta$, then

$$|F(C_s)|^2 = \rho^2 d^2 + w^2.$$

Hence

$$|F(C_s)|^2 - |C_s|_0^2 = \rho^2 d^2 + w^2 - (d^2 + 1) = (w^2 - 1) - (1 - \rho^2)d^2.$$

Since $|d| \leq \beta$,

$$(1 - \rho^2)d^2 \leq (1 - \rho^2)\beta^2 = \beta^2 - \lambda^2,$$

so

$$|F(C_s)|^2 - |C_s|_0^2 \geq w^2 - 1 - (\beta^2 - \lambda^2) > 0. \quad (3)$$

Thus every pulled-back bend is strictly lengthened by F .

Choose finitely many bends $C_{s_0}, \dots, C_{s_{N-1}}$ in cyclic order, with mesh sufficiently small. Consecutive bends bound clean quadrilaterals in M_β . Split each such quadrilateral by one of its diagonals. This gives a finite clean triangulation of M_β . Define a piecewise-affine map $\bar{f}$ by setting $\bar{f} = F$ on all vertices and extending affinely over each triangle.

For sufficiently small mesh, $\bar{f}$ is an embedding.

Reviewer 1

I have some question about the C^1 -convergence. Technically the PL map is not C^1 in the ordinary sense. However, this fine point seems like it can be easily tightened up.

Indeed, in bend coordinates the smooth map F has the form

$$F(s, r) = (1 - r)a(s) + rb(s),$$

with a, b C^1 boundary curves. The above piecewise-affine maps are the usual secant approximations to this ruled embedding; they converge to F in C^1 . A C^1 -small perturbation of a compact C^1 -embedded surface is again embedded.

Now check the squeeze inequalities. If e is a boundary edge of flat length ℓ , then the boundary arc of F between its endpoints has length $\rho\ell$, so the chord $\bar{f}(e)$ has length at most $\rho\ell < \ell$. Hence boundary edges are strictly shortened.

I like that this argument is conceptual and does not get bogged down in too much fine-scale detail.

The nonboundary edges are of two types: selected bends and diagonals between two sufficiently close selected bends. Selected bends are strictly lengthened by (3). For the diagonals, the relevant squared length difference depends continuously on the two adjacent bend parameters and agrees, when the parameters coincide, with the positive quantity in (3). Compactness of the parameter circle gives a uniform positive margin, so for sufficiently small mesh every diagonal is also strictly lengthened.

Therefore $\bar{f}$ restricts to a squeeze map on every triangle of the clean triangulation of M_β . Lifting the triangulation and the map to Σ gives a G_β -invariant clean triangulation and a continuous map

$$f : \Sigma \rightarrow \mathbb{R}^3$$

with $f = \bar{f} \circ \pi$, where $\pi : \Sigma \rightarrow M_\beta$ is the quotient map. Since $\bar{f}$ is an embedding,

$$f(p) = f(q) \iff \pi(p) = \pi(q) \iff p = G_\beta^k(q) \text{ for some } k \in \mathbb{Z}.$$

Thus β is realized. □

Theorem 7. *The set of realized numbers is exactly $(\sqrt{3}, \infty)$. In particular, its infimum is $\sqrt{3}$.*

Proof. The lower-bound lemma shows that every realized β satisfies $\beta > \sqrt{3}$. The proposition shows that every $\beta > \sqrt{3}$ is realized. Hence the set of realized numbers is $(\sqrt{3}, \infty)$, whose infimum is $\sqrt{3}$. □

References

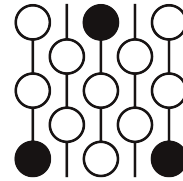
- [1] M. Sadowsky, *Ein elementarer Beweis fuer die Existenz eines abwickelbaren Moebiuschen Bandes und die Zurueckfuehrung des geometrischen Problems auf ein Variationsproblem*, Sitzungsberichte der Preussischen Akademie der Wissenschaften, physikalisch-mathematische Klasse **22** (1930), 412–415.
- [2] D. F. Hinz and E. Fried, *Translation of Michael Sadowsky’s paper “An elementary proof for the existence of a developable Moebius band and the attribution of the geometric problem to a variational problem”*, J. Elasticity **119** (2015), 3–6.
- [3] R. E. Schwartz, *The optimal paper Moebius band*, Ann. of Math. (2) **201** (2025), no. 1, 291–305.

Review of Submission 2C

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. Similar to Submission B, this solution provides a correct argument both for the lower bound $\beta \geq \sqrt{3}$ and for the realizability of every $\beta > \sqrt{3}$. It accomplishes the first goal by adapting Schwartz’s argument, which shows that a smoothly isometrically embedded Möbius strip has aspect ratio greater than $\sqrt{3}$, to the PL setting in which the triangles in the triangulation satisfy the “squeeze map” hypothesis. This adaptation is carried out without any major logical flaws.

For the second goal, the solution starts with the piecewise smooth ruled isometric embeddings of the Möbius band due to Sadowsky and Halpern–Weaver, approximates these embeddings by PL embeddings, and then carefully chooses scaling factors for the domain Möbius band so that the resulting PL embedding satisfies the squeeze property. This part of the argument is also supported by justifications, including detailed inequalities, and a verification that the chosen scaling achieves the squeeze property on each triangle. However, the argument involving embeddedness of PL approximations of the smooth embedding is confusing. See the inline comment for Proposition 6. Nevertheless, I believe that the statement is salvagable using perhaps a slightly different strategy, such as working directly with the explicit construction.

Novelty. Similar to Submission B, for both parts of the argument, the strategies implemented by this solution are essentially the same as the expected strategy for this problem, the strategy followed by the human solution, and the strategies pursued by some of the other AI models. Therefore, this approach should not be considered novel.

For the first part, the solution identifies that the proof strategy underlying Schwartz’s theorem can be modified to prove the lower bound $\beta > \sqrt{3}$ in the setting of the problem statement. However, given access to Schwartz’s result, adapting the argument to the PL category is a relatively routine task. In particular, Lemmas 4 and 5 use proof strategies that are very similar to those of corresponding lemmas in Schwartz’s paper. The solution however fails to cite Schwartz’s work explicitly in this part of the argument.

For the second part, the solution uses PL approximations of a known construction in the smooth setting. This is also a standard strategy, provided the necessary background from the existing literature is available.

2 Recommendation

I consider this solution to be publishable after minor revisions. There are some minor errors in the statement of Lemma 3(i), but these do not affect the overall proof. There are also some inaccuracies in the discussion of the relevant literature, but these can likewise be corrected.

3 Submission with inline comments

Abstract

We prove the stronger statement that a number $\beta > 0$ is realized in the sense of the question if and only if $\beta > \sqrt{3}$. Hence the infimum of the set of realized numbers is $\sqrt{3}$.

3.1 Preliminaries

Let

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$$s(x, 0) = x \pmod{2\beta}, \quad s(x, 1) = x + \beta \pmod{2\beta}.$$

Lemma 1 (quotient and boundary contraction). *Suppose β is realized. Then the map f descends to a topological embedding*

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Moreover, the G_β -invariant clean triangulation descends to a finite clean triangulation of M_β , and every nondegenerate boundary arc $A \subset \partial M_\beta$ satisfies

$$\text{length}(\bar{f}(A)) < \text{length}_0(A),$$

where length_0 denotes flat boundary length.

Proof. Condition (2) gives $f \circ G_\beta = f$, so f descends to $\bar{f}$. The same condition says precisely that $\bar{f}$ is injective. Since M_β is compact and $\mathbb{R}^3$ is Hausdorff, $\bar{f}$ is an embedding.

A compact fundamental domain in Σ meets only finitely many triangles by local finiteness, so only finitely many G_β -orbits of triangles occur. Thus the quotient triangulation is finite.

Finally, on each boundary edge of the triangulation, f is affine and strictly shortens that edge. Hence it strictly shortens every subsegment of every boundary edge. Since every boundary arc is a finite union of such subsegments, its image has strictly smaller length. $\square$

Lemma 2 (the squeeze cone). *Let τ be a clean triangle with boundary edge AB and opposite vertex C . If an affine map $L : \tau \rightarrow \mathbb{R}^3$ decreases the length of AB and increases the lengths of AC and BC , then L increases the length of every segment joining C to a point of AB .*

Proof. Write $u = C - A$, $v = C - B$, so $u - v = B - A$. Let

$$q(w) = |Lw|^2 - |w|^2.$$

The hypotheses are

$$q(u) > 0, \quad q(v) > 0, \quad q(u - v) < 0.$$

For $0 \leq t \leq 1$,

$$q((1 - t)u + tv) = (1 - t)q(u) + tq(v) - t(1 - t)q(u - v) > 0.$$

Thus every cevian from C to the base AB is strictly lengthened. □

3.2 The canonical ruling and a T -pattern

For every clean triangle, foliate it by the straight segments joining the vertex opposite the boundary edge to points of the boundary edge. Adjacent clean triangles share a nonboundary edge, and that edge is one of the leaves in both triangles. Therefore these local foliations glue.

Lemma 3 (canonical ruling). *The quotient clean triangulation of M_β carries a continuous circle-family $\mathcal{L} = \{L_t\}_{t \in S^1}$ of straight clean segments with the following properties.*

- (i) *Each L_t is contained in a single clean triangle and joins the two boundary components in the universal cover.*
- (ii) *The relative interiors of distinct leaves are disjoint and the leaves cover M_β .*
- (iii) *The image $\bar{f}(L_t)$ is a straight segment in $\mathbb{R}^3$, and*

$$|\bar{f}(L_t)| > |L_t|_0 \geq 1.$$

- (iv) *A continuous orientation of the leaves reverses after one circuit of the parameter circle.*

Reviewer 2

(Minor comment) Lemma 3(i) is not true as stated. Some of the L_t 's are boundary edges shared by two adjacent clean triangles, and hence are contained in both clean triangles. This issue can nevertheless be resolved by modifying condition (i) to account for this edge case. Moreover, it does not appear to affect the later argument that invokes Lemma 3.

Proof. Let $\mathcal{T}$ be the finite clean triangulation of M_β . Form the dual graph whose vertices are triangles and whose edges correspond to shared nonboundary edges. Every triangle has exactly two nonboundary edges, and every nonboundary edge is shared by two triangles, so the dual graph is 2-regular. It is connected: at each boundary vertex the link is an interval, so the triangles incident to that vertex are connected through shared nonboundary edges; since the triangulation is connected, the dual graph is connected. Hence the dual graph is a single cycle.

On a triangle τ , the family of cevians from the opposite vertex to the boundary edge is parametrized by an interval; its two endpoint leaves are the two nonboundary edges of τ . Gluing

these intervals according to the cyclic dual graph gives a parameter circle S^1 . This proves the existence and continuity of the family $\{L_t\}$, and the relative interiors are plainly disjoint.

Each leaf is contained in a clean triangle, and by the preceding squeeze-cone lemma its image is strictly longer than its flat length. Since every clean segment crosses from $y = 0$ to $y = 1$ in the universal cover, its flat length is at least 1.

Finally, the leaf family is an interval-bundle structure on the Mobius band away from finitely many boundary vertices. Since M_β is the nontrivial I -bundle over a circle, the orientation line of the fibers is nontrivial. Equivalently, in the universal cover, going once around the leaf circle applies the deck transformation G_β , which exchanges the two boundary components. Hence any continuous orientation of the leaves is reversed after one circuit. $\square$

Lemma 4 (*T-pattern*). *There are two distinct leaves L_s, L_t such that the two straight segments*

$$\bar{f}(L_s), \quad \bar{f}(L_t)$$

lie in perpendicular coplanar lines.

Proof. Lift the parameter circle to $\mathbb{R}$. Choose continuously oriented unit direction vectors $u(r)$ for the image segments and let $m(r)$ denote their midpoints. By the orientation-reversal property,

$$u(r + 2\pi) = -u(r), \quad m(r + 2\pi) = m(r).$$

Let

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Compactify Υ by adding two points: ∂_+ , corresponding to $s \rightarrow r$, and ∂_- , corresponding to $s \rightarrow r + 2\pi$. This compactification is a 2-sphere, and the swap involution

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extends to the antipodal map on that sphere.

Define

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Then F extends continuously to the compactification with

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and satisfies

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By the Borsuk–Ulam theorem, F has a zero. At such a zero, the two direction vectors are perpendicular, and the vector joining the midpoints is orthogonal to the normal $u(r) \times u(s)$. Therefore the two supporting lines are coplanar. This gives the desired pair of leaves. $\square$

3.3 The lower bound

Lemma 5 (geometry of a squeezed T -pattern). *If β is realized, then $\beta > \sqrt{3}$.*

Proof. Let T and B be the two leaves supplied by the preceding lemma. Write

$$T^* = \bar{f}(T), \quad B^* = \bar{f}(B).$$

Their supporting lines are perpendicular and coplanar. Interchange T and B , if necessary, so that the line supporting T^* does not meet the relative interior of B^* . Otherwise the two embedded segments would have an interior intersection.

Let u be the endpoint of B whose image is farthest from the line supporting T^* . Since B^* is perpendicular to that line and the line does not meet the relative interior of B^* , the distance h from $\bar{f}(u)$ to the line supporting T^* satisfies

$$h \geq |B^*| > |B|_0 \geq 1.$$

Let the endpoints of T divide ∂M_β into two boundary arcs H and D , with $u \in D$. Choose the signed horizontal displacement t of T so that

$$\text{length}_0(H) = \beta - t, \quad \text{length}_0(D) = \beta + t.$$

Then

$$|T|_0 = \sqrt{1 + t^2}.$$

Because T is a leaf, it is lengthened:

$$|T^*| > \sqrt{1 + t^2}.$$

Since H is a boundary path joining the endpoints of T ,

$$\sqrt{1 + t^2} < |T^*| \leq \text{length}(\bar{f}(H)) < \beta - t. \quad (1)$$

Now consider the planar triangle Δ whose base is T^* and whose third vertex is $\bar{f}(u)$. Its base has length $> |T|_0 = \sqrt{1 + t^2}$, and its height is $h > 1$. If $\vee$ denotes the union of the two nonbase sides of Δ , the elementary reflection argument gives

$$\text{length}(\vee) > \sqrt{5 + t^2}.$$

Indeed, among all triangles with base length b and height h , the sum of the two nonbase sides is at least $\sqrt{b^2 + 4h^2}$.

The boundary arc D runs from one endpoint of T to the other through u . Hence the two sides of $\vee$ are no longer than the two corresponding subarcs of $\bar{f}(D)$. Therefore

$$\sqrt{5 + t^2} < \text{length}(\vee) \leq \text{length}(\bar{f}(D)) < \beta + t. \quad (2)$$

From (1) and (2),

$$2\beta > \alpha(t) := \sqrt{1 + t^2} + \sqrt{5 + t^2},$$

and also

$$2\beta > \gamma(t) := 2\sqrt{5+t^2} - 2t.$$

Let $t_0 = 1/\sqrt{3}$. Then

$$\alpha(t_0) = \gamma(t_0) = 2\sqrt{3}.$$

Moreover, α is increasing for $t > 0$, and γ is strictly decreasing on $\mathbb{R}$. Thus for every real t ,

$$\max\{\alpha(t), \gamma(t)\} \geq 2\sqrt{3}.$$

Consequently $2\beta > 2\sqrt{3}$, so $\beta > \sqrt{3}$. □

Reviewer 2

(Minor comment) The proofs of Lemmas 4 and 5 appear to implement essentially the same strategies as those given or outlined in Schwartz’s paper for similar lemmas. The solution should therefore cite Schwartz’s work explicitly in this part of the argument, rather than presenting these proof strategies as if they were novel.

3.4 Realization for every $\beta > \sqrt{3}$

We use the classical existence theorem for paper Mobius bands: for every width $w > 0$ and every length λ with $\lambda/w > \sqrt{3}$, there exists a smooth embedded isometric Mobius band of width w and length λ , obtained by smoothing the triangular Mobius band. Moreover, it has a continuous ruling by straight bends whose preimages are straight segments joining the two boundary components. This is the standard Sadowsky–Wunderlich construction; see Sadowsky [1], the translation by Hinz–Fried [2], and the modern discussion in Schwartz [3].

Reviewer 2

(Major comment) The solution claims that there exists a “smooth embedded isometric Mobius band...” and cites Sadowsky’s 1930 work, together with a modern English translation by Hinz–Fried. However, the isometric embedding constructed by Sadowsky is not smooth; rather, it is piecewise smooth and globally C^1 . The first construction of a globally smooth isometric embedding is due to Halpern–Weaver in their 1977 paper *Inverting a cylinder through isometric immersions and embeddings*. This misattribution suggests that the solution does not accurately reflect the relevant literature.

Proposition 6. *Every $\beta > \sqrt{3}$ is realized.*

Proof. Fix $\beta > \sqrt{3}$. Choose numbers w, λ such that

$$1 < w, \quad \sqrt{3}w < \lambda < \beta, \quad w^2 - 1 > \beta^2 - \lambda^2.$$

This is possible by taking λ sufficiently close to β and then $w > 1$ sufficiently close to 1.

Let $\rho = \lambda/\beta < 1$. Let

$$P_{\lambda,w} = ([0, \lambda] \times [0, w]) / (0, y) \sim (\lambda, w - y),$$

and let

$$I : P_{\lambda,w} \longrightarrow \mathbb{R}^3$$

be a smooth embedded paper Mobius band with a ruling by straight bends. Define

$$A : M_\beta \longrightarrow P_{\lambda,w}, \quad A[x, y] = [\rho x, wy].$$

This is well-defined because

$$A(x + \beta, 1 - y) = (\rho x + \lambda, w - wy),$$

which is the defining Mobius identification in $P_{\lambda,w}$. Put

$$F = I \circ A : M_\beta \longrightarrow \mathbb{R}^3.$$

Then F is an embedding.

Let $\{C_s\}_{s \in S^1}$ be the pulled-back bend ruling on M_β . Each C_s is a clean straight segment. If C_s has signed horizontal displacement d , with $|d| \leq \beta$, then

$$|F(C_s)|^2 = \rho^2 d^2 + w^2.$$

Hence

$$|F(C_s)|^2 - |C_s|_0^2 = \rho^2 d^2 + w^2 - (d^2 + 1) = (w^2 - 1) - (1 - \rho^2)d^2.$$

Since $|d| \leq \beta$,

$$(1 - \rho^2)d^2 \leq (1 - \rho^2)\beta^2 = \beta^2 - \lambda^2,$$

so

$$|F(C_s)|^2 - |C_s|_0^2 \geq w^2 - 1 - (\beta^2 - \lambda^2) > 0. \quad (3)$$

Thus every pulled-back bend is strictly lengthened by F .

Choose finitely many bends $C_{s_0}, \dots, C_{s_{N-1}}$ in cyclic order, with mesh sufficiently small. Consecutive bends bound clean quadrilaterals in M_β . Split each such quadrilateral by one of its diagonals. This gives a finite clean triangulation of M_β . Define a piecewise-affine map $\bar{f}$ by setting $\bar{f} = F$ on all vertices and extending affinely over each triangle.

For sufficiently small mesh, $\bar{f}$ is an embedding. Indeed, in bend coordinates the smooth map F has the form

$$F(s, r) = (1 - r)a(s) + rb(s),$$

with a, b C^1 boundary curves. The above piecewise-affine maps are the usual secant approximations to this ruled embedding; they converge to F in C^1 . A C^1 -small perturbation of a compact C^1 -embedded surface is again embedded.

Reviewer 2

(Major comment) The argument that the secant approximations converge in C^1 is confusing as written. First, the PL embeddings themselves are only piecewise C^1 , rather than globally C^1 , so the phrase “ C^1 embedding” is not appropriate in this context without further clarification. I believe that the intended statement is likely true, but it requires a much clearer argument. One possible approach would be to prove embeddedness directly from the explicit Sadowsky construction using the geometry of planar equilateral triangles and half-cylinders.

Now check the squeeze inequalities. If e is a boundary edge of flat length ℓ , then the boundary arc of F between its endpoints has length $\rho\ell$, so the chord $\bar{f}(e)$ has length at most $\rho\ell < \ell$. Hence boundary edges are strictly shortened.

The nonboundary edges are of two types: selected bends and diagonals between two sufficiently close selected bends. Selected bends are strictly lengthened by (3). For the diagonals, the relevant squared length difference depends continuously on the two adjacent bend parameters and agrees, when the parameters coincide, with the positive quantity in (3). Compactness of the parameter circle gives a uniform positive margin, so for sufficiently small mesh every diagonal is also strictly lengthened.

Reviewer 2

Not sure how much the argument here for diagonal stretching needs to be elaborated, but I nevertheless believe that the statement is still true.

Therefore $\bar{f}$ restricts to a squeeze map on every triangle of the clean triangulation of M_β . Lifting the triangulation and the map to Σ gives a G_β -invariant clean triangulation and a continuous map

$$f : \Sigma \rightarrow \mathbb{R}^3$$

with $f = \bar{f} \circ \pi$, where $\pi : \Sigma \rightarrow M_\beta$ is the quotient map. Since $\bar{f}$ is an embedding,

$$f(p) = f(q) \iff \pi(p) = \pi(q) \iff p = G_\beta^k(q) \text{ for some } k \in \mathbb{Z}.$$

Thus β is realized. □

Theorem 7. *The set of realized numbers is exactly $(\sqrt{3}, \infty)$. In particular, its infimum is $\sqrt{3}$.*

Proof. The lower-bound lemma shows that every realized β satisfies $\beta > \sqrt{3}$. The proposition shows that every $\beta > \sqrt{3}$ is realized. Hence the set of realized numbers is $(\sqrt{3}, \infty)$, whose infimum is $\sqrt{3}$. □

References

- [1] M. Sadowsky, *Ein elementarer Beweis fuer die Existenz eines abwickelbaren Moebiuschen Bandes und die Zurueckfuehrung des geometrischen Problems auf ein Variationsproblem*, Sitzungsberichte der Preussischen Akademie der Wissenschaften, physikalisch-mathematische Klasse **22** (1930), 412–415.
- [2] D. F. Hinz and E. Fried, *Translation of Michael Sadowsky’s paper “An elementary proof for the existence of a developable Moebius band and the attribution of the geometric problem to a variational problem”*, J. Elasticity **119** (2015), 3–6.
- [3] R. E. Schwartz, *The optimal paper Moebius band*, Ann. of Math. (2) **201** (2025), no. 1, 291–305.